

MATHIAS BERTORELLI ARGIBAY

MLOps · ML Engineer · GenAI Engineer

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Computer Science degree, MSc in Data Science. I design and run the platforms ML teams build on — with a strong focus on good practices and reliable, scalable solution design.

• EXPERIENCE

Computer Vision Engineer — Freelance

Oct 2024 – Present

Innovision

- Ongoing freelance collaboration: vision systems for industrial production lines, delivered end to end as sole developer.
- Real-time mesh defect detection: UNet segmentation on an industrial camera feed with physical measurements overlaid live; high-precision screen edge detection; bakery-line defect detection.

Python · PyTorch · OpenCV · Industrial cameras · Docker

Associate Director — MLOps

Nov 2024 – Present

AstraZeneca – Alexion

- Designed and built a multi-tenant AWS cloud platform — one account per environment (dev, uat, preprod, prod) with isolated project tenants — leading a team of engineers; now product owner of the platform.
- Self-service infrastructure: curated Terraform templates let development teams provision their own compliant resources, removing cross-team bottlenecks.
- Lead internal and vendor teams across the full project lifespan, from kickoff to productionalization; project blueprints covering solution design, architecture, cybersecurity and compliance.
- Led and contributed to agentic frameworks and LLM-based solutions.

AWS · Terraform · Platform engineering · GenAI

Machine Learning Engineer

Sep 2023 – Nov 2024

ISDIN

- Owned the full ML model lifecycle on Google Cloud Platform, end to end.
- Defined good practices and methodologies for the Data Science team; trained Data Scientists in tooling and practices.
- Designed and maintained internal Python packages that sped up new DS projects.
- Designed and implemented the Feature Store and the MLOps / CI-CD pipelines used by the team.
- LLM projects: prompt engineering and RAG.

GCP · Terraform · Airflow · Docker · PaLM

Machine Learning Engineer

Jan 2023 – Sep 2023

Mad Collective

- Maintained and improved the cloud infrastructure used by the entire Data Science team.
- Refactored legacy code; ensured DS repositories ship with tests and Docker containers.
- Maintained in-house Python packages on an internal DevPI.

Terraform · AWS (EC2, ECS, VPC, S3) · Docker · Airflow

Machine Learning Engineer

Jun 2021 – Nov 2022

Vintra

- Built a distributed system generating image batches to improve model training, using vector databases and MySQL.
- Exported neural-network models to ONNX and TensorRT, with validation of correct operation in the new format.
- Built the proof of concept that introduced the MLOps methodology (Apache Airflow) to the company workflow.

TensorFlow · PyTorch · FAISS · OpenCV · TensorRT · ONNX · Airflow

Researcher / Backend Developer — inLab FIB

Dec 2018 – Jun 2021

UPC, Barcelona

- ML, deep learning and computer vision projects with SEAT and Cdmon: computer vision, churn prediction, forecasting and NLP.
- Backend development for external clients (Django, Flask, Docker).

- **EDUCATION**

MSc in Data Science — La Salle (2022–2023)

Degree in Informatics Engineering — UPC, FIB (2015–2021)

- **ADDITIONAL**

Languages: Spanish and Catalan (native) · English (advanced)

Core stack: Python · Terraform · Docker · Airflow · GCP · AWS · CI/CD · MLOps · RAG / agentic systems